



Department of Computer Science and Engineering (Data Science)

B.Tech. Sem: III Subject: Statistics for Data Science

Experiment 2

Name: Bhuvi Ghosh
60009210191

SAP ID:

Date:	Experiment Title: Measures of Central tendency and dispersion
Aim	To measure central tendency and dispersion of data using Python
Software	Google Colab
Implementation	1. Find arithmetic mean of 20, 2, 7, 1, 34 <pre> import numpy as np import statistics as stats from scipy import stats from scipy.stats import gmean,hmean import matplotlib.pyplot as plt a=[20,2,7,1,34] np.mean(a) </pre> 12.8 2. Create following matrix using multidimensional array and calculate arithmetic mean for each column, each row and considering entire data. <pre> 14 17 12 33 44 15 6 27 8 19 23 2 54 1 4 </pre> <pre> [4] a=[[14,17,12,13,44],[15,6,27,8,19],[23,2,54,1,4]] z1=np.mean(a) print(z1) z2=np.mean(a,axis=0) print(z2) z3=np.mean(a,axis=1) print(z3) </pre> <pre> 17.266666666666666 [17.33333333 8.33333333 31. 7.33333333 22.33333333] [20. 15. 16.8] </pre> 3. Find minimum value, maximum value and range for entire data, column wise and row wise. <pre> 3,7,5 8,4,3 2,4,9 </pre>

```

[5] z=[[3,7,5],[8,4,3],[2,4,9]]
print(np.min(z))
print(np.max(z))
print(np.ptp(z))
print(np.min(z,axis=0))
print(np.max(z,axis=1))
print(np.ptp(z,axis=0))
print(np.ptp(z,axis=1))

2
9
7
[2 4 3]
[7 8 9]
[6 3 6]
[4 5 7]

```

4. Find weighted average for the data given below.

Outcomes	Frequency
1	4
2	3
3	2
4	1

```

[6] z1=[1,2,3,4]
z2=[4,3,2,1]
print(np.average(z1,weights=z2))

2.0

```

5. The speed of 13 vehicles is 99,86,87,88,111,86,103,87,94,78,77,85,86
 Find mean, median and mode.

```

[7] z1=[99,86,87,88,111,86,103,87,94,78,77,85,86]
z2=np.mean(z1)
print(z2)
z3=np.median(z1)
print(z3)
z4=stats.mode(z1)
print(z4)

89.76923076923077
87.0
ModeResult(mode=array([86]), count=array([3]))

```

6. Calculate geometric mean for each column, each row and considering entire data.

1 3 27
 3 4 6
 7 6 3
 3 6 8

```
{x}
[8] z1=[[1,3,27],[3,4,6],[7,6,3],[3,6,8]]
    z2=gmean(z1,axis=0)
    print(z2)
    z3=gmean(z1,axis=1)
    print(z3)
    z4=gmean(z1)
    print(z4)

[2.81731325 4.55901411 7.89644408]
[4.32674871 4.16016765 5.01329793 5.24148279]
[2.81731325 4.55901411 7.89644408]
```

7. Calculate harmonic mean for 1, 3, 5, 7, 9

```
<> [9] z=[1,3,5,7,9]
    print(hmean(z))

2.797513321492007
```

8. Calculate median for each column, each row and considering entire data.

30 65 70

80 95 10

50 90 60

```
z3=[[30,65,70],[80,95,10],[50,90,60]]
print(np.median(z3,axis=0))
z2=np.median(z3,axis=1)
print(z2)
np.median(z3)
```

```
[50. 90. 60.]
[65. 80. 60.]
65.0
```

9. The number of solar heating systems available to the public is quite large, and their heat-storage capacities are quite varied. Here is a distribution of heat-storage capacity (in days) of 28 systems that were tested recently by University Laboratories, Inc.:

Days	Frequency
0-0.99	2
1-1.99	4
2-2.99	6
3-3.99	7
4-4.99	5
5-5.99	3
6-6.99	1

University Laboratories, Inc., knows that its report on the tests will be widely circulated and used as the basis for tax legislation on solar-heat allowances. It therefore wants the measures it uses to be as reflective of the data as possible.

(a) Compute the mean for these data.

```

a = [[0,0.99,2],[1,1.99,4],[2,2.99,6],[3,3.99,7],[4,4.99,5],[5,5.99,3],[6,6.99,1]]
n = 0
d = 0

a[6][1] = (a[6][1]+7)/2
for i in range (0,6):
    a[i][1] = (a[i][1] + a[i+1][0])/2
    a[i+1][0] = a[i][1]

for i in range(0,7):
    mi = (a[i][0]+a[i][1])/2
    n = n + a[i][2]*mi
    d = d + a[i][2]

mean = n/d
print(mean)

```

3.280892857142857

(b) Compute the mode for these data.

```

[71] a = [[0,0.99,2],[1,1.99,4],[2,2.99,6],[3,3.99,7],[4,4.99,5],[5,5.99,3],[6,6.99,1]]
a[6][1] = (a[6][1]+7)/2
for i in range (0,6):
    a[i][1] = (a[i][1] + a[i+1][0])/2
    a[i+1][0] = a[i][1]

max = -1;
for i in range (0,7):
    if(a[i][2]>max):
        max = a[i][2]
        index = i

f0 = a[index-1][2]
f1 = a[index][2]
f2 = a[index+1][2]
L = a[index][0]
I = (a[index][1]-a[index][0])

mode = L + ((f1-f0)/(2*f1-f0-f2))*I
print(mode)

```

3.3283333333333336

(c) Compute the median for these data.

```

a = [[0,0.99,2],[1,1.99,4],[2,2.99,6],[3,3.99,7],[4,4.99,5],[5,5.99,3],[6,6.99,1]]

a[6][1] = (a[6][1]+7)/2
for i in range(0,6):
    a[i][1] = (a[i][1] + a[i+1][0])/2
    a[i+1][0] = a[i][1]

cf = 0
for i in range(0,7):
    cf = cf + a[i][2]
    a[i].append(cf)

N = a[6][3]

for i in range(0,7):
    if(a[i][3]>N/2):
        index = i
        break

L = a[index][0]
cf = a[index-1][3]
f = a[index][2]
I = (a[index][1]-a[index][0])

median = L + (((N/2)-cf)/f)*I
print(median)

```

3.2807142857142857

(d) Select the answer among parts (a), (b), and (c) that best reflects the central tendency of the test data and justify your choice.

Median is the best measure of central tendencies as it is more reliable & it Does not take into consideration the outliers. Also it can be used for a more varied frequency range.

10. Write a function for calculating percentile and determine 30, 50, 75 and 90 percentiles for the following data.

30,40,72,83,25,10,50,90,60,15,5,9,34,23,67,80,67,45

```

[10] z=[30,40,72,83,25,10,50,90,60,15,5,9,34,23,67,80,67,45]
print(np.percentile(z,30))
print(np.percentile(z,50))
print(np.percentile(z,75))
print(np.percentile(z,90))

25.5
42.5
67.0
80.9

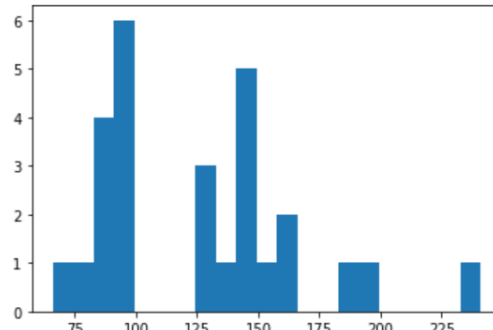
```

11. The numbers of apartments in 27 apartment complexes in Cary, North Carolina, are given below. 91 79 66 98 127 139 154 147 192 88 97 92 87 142 127 184 145 162 95 89 86 98 145 129 149 158 241

(a) Construct a frequency distribution using intervals 66–87, 88–109, 220–241.

```
z=[91,79,66,98,127,139,154,147,192,88,97,92,87,142,127,184,145,162,95,89,86,98,145,129,149]
plt.hist(z,21,range=(np.min(z),np.max(z)))
z1=stats.mode(z)
print(z1)
np.mean(z)
```

```
ModeResult(mode=array([98]), count=array([2]))
126.18518518518519
```



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(b) Estimate the modal value.

```
z1=stats.mode(z)
print(z1)
```

```
ModeResult(mode=array([1]), count=array([1]))
```

(c) Compute the mean of the raw data.

```
np.mean(z)
```

```
5.0
```

(d) Compare your answers in parts (b) and (c) and comment on which of the two is the better measure of central tendency of these data and why.

Mean is a better measure of central tendency as it gives us the average of all possible occurring events which can be used efficiently for analysis.

12. There are 39 plants in the garden. A few plants were selected randomly and their heights in cm were recorded as follows: 51, 38, 79, 46, 57. Calculate the standard deviation of their heights.

```
z1=[51,38,79,46,57]
print(np.std(z1))
```

```
13.876599006961325
```

13. The Casual Life Insurance Company is considering purchasing a new fleet of company cars. The financial department's director, Tom Dawkins, sampled 40 employees to determine the number of miles each drove over a 1-year period. The results of the study follow. Calculate the range and interquartile range.

3,600	4,200	4,700	4,900	5,300	5,700	6,700	7,300
7,700	8,100	8,300	8,400	8,700	8,700	8,900	9,300
9,500	9,500	9,700	10,000	10,300	10,500	10,700	10,800
11,000	11,300	11,300	11,800	12,100	12,700	12,900	13,100
13,500	13,800	14,600	14,900	16,300	17,200	18,500	20,300

```
[22] z1=[3600,4200,4700,4900,5300,5700,6700,7300,7700,8100,8300,8400,8700,8700,8900,9300,9500,9700,10000,10300,10500,10700,10800,11000,11300,11300,11800,12100,12700,12900,13100,13500,13800,14600,14900,16300,17200,18500,20300]
range=np.max(z1)-np.min(z1)
print(range)
interquartilerange=np.percentile(z1,75)-np.percentile(z1,25)
print(interquartilerange)

16700
4600.0
```

14. The head chef of The Flying Taco has just received two dozen tomatoes from her supplier, but she isn't ready to accept them. She knows from the invoice that the average weight of a tomato is 7.5 ounces, but she insists that all be of uniform weight. She will accept them only if the average weight is 7.5 ounces and the standard deviation is less than 0.5 ounce. Here are the weights of the tomatoes

6.3 7.2 7.3 8.1 7.8 6.8 7.5 7.8 7.2 7.5 8.1 8.2
 8.0 7.4 7.6 7.7 7.6 7.4 7.5 8.4 7.4 7.6 6.2 7.4

What is the chef's decision and why?

```
[24] z1=[6.3,7.2,7.3,8.1,7.8,6.8,7.5,7.8,7.2,7.5,8.1,8.2,8,7.4,7.7,7.6,7.4,7.5,8.4,7.4,7.6,6.2,7.4]
print(np.mean(z1))
print(np.std(z1))

7.495652173913043
0.5270741725457995
```

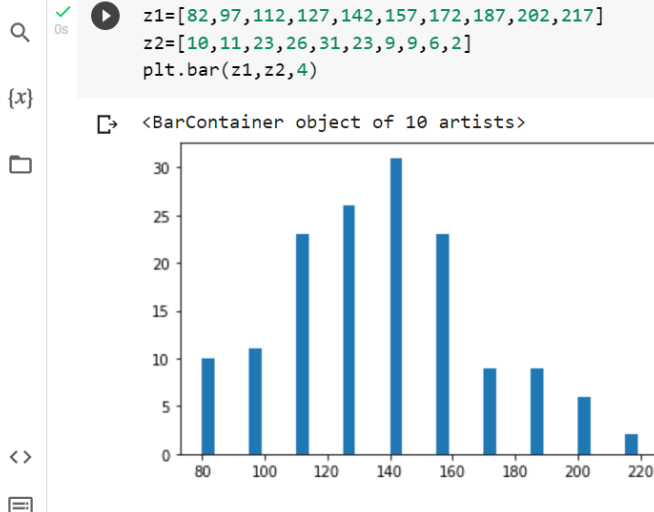
The chef will not accept the tomatoes as the average weight is lesser than 7.5 & standard deviation is less than 0.5.

15. A company is considering employing one of two training programs. Two groups were trained for the same task. Group 1 was trained by program A; group 2, by program B. For the first group, the times required to train the employees had an average of 32.11 hours and a variance of 68.09. In the second group, the average was 19.75 hours and the variance was 71.14. Which training program has less relative variability in its performance?

Program A has lower variability as variance is less.

16. Here is a frequency distribution of the weight of 150 people who used a ski lift a certain day. Construct a histogram for these data.

Class	Frequency	Class	Frequency
75-89	10	150-164	23
90-104	11	165-179	9
105-119	23	180-194	9
120-134	26	195-209	6
135-149	31	210-224	2



(a) What can you see from the histogram about the data that was not immediately apparent from the frequency distribution?

(b) If each ski lift chair holds two people but is limited in total safe weight capacity to 400 pounds, what can the operator do to maximize the people capacity of the ski lift without exceeding the safe weight capacity of a chair? Do the data support your proposal?

More people from the weight category 75-89kg & 90-104kg can be accommodated to increase the capacity of lift.

17. Test scores for a college statistics class held during the day are:

99 56 78 55.5 32 90 80 81 56 59 45 77 84.5 84 70 72 68 32 79 90

Test scores for a college statistics class held during the evening are:

98 78 68 83 81 89 88 76 65 45 98 90 80 84.5 85 79 78 98 90 79 81 25.5

- a. Find the smallest and largest values, the median, and the first and third quartile for the day class.

{x}

✓ 0s

```
[30] z1=[99,56,78,55.5,32,90,80,81,56,59,45,77,84.5,84,70,72,68,32,79,90]
print(np.min(z1))
print(np.max(z1))
print(np.median(z1))
print(np.percentile(z1,25))
print(np.percentile(z1,75))
```

32.0
 99.0
 74.5
 56.0
 81.75

- b. Find the smallest and largest values, the median, and the first and third quartile for the night class.

```
[32] z3=[98,78,68,83,81,89,88,76,65,45,98,90,80,84.5,85,79,78,98,90,79,81,25.5]
print(np.min(z3))
print(np.max(z3))
print(np.median(z3))
print(np.percentile(z3,25))
print(np.percentile(z3,75))
```

25.5
 98.0
 81.0
 78.0
 88.75

- c. For each data set, what percentage of the data is between the smallest value and the first quartile? the first quartile and the median? the median and the third quartile? the third quartile and the largest value? What percentage of the data is between the first quartile and the largest value?

Smallest value & first quartile:25% data

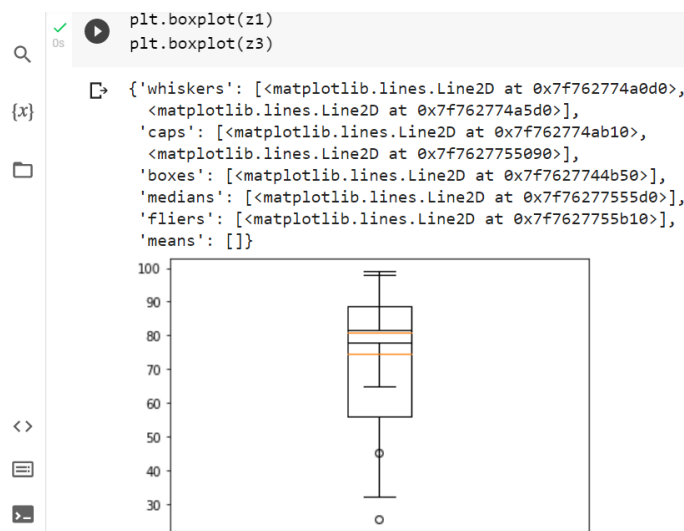
First quartile & median:25% data

Median & third quartile:25% data

Third quartile & largest value:25% data

Smallest value & largest value:100% data

- d. Create a box plot for each set of data. Use one number line for both box plots.



- e. Which box plot has the widest spread for the middle 50% of the data (the data between the first and third quartiles)? What does this mean for that set of data in comparison to the other set of data.



	Test scores during the day have a wider score in the middle 50% as the interquartile range is more. This means that more number of people in the day shift have scores between 25% & 75%. <pre>[38] np.percentile(z1,75)-np.percentile(z1,25)</pre> 25.75 <pre>np.percentile(z3,75)-np.percentile(z3,25)</pre> 10.75
Conclusion	

Signature of Faculty